

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 3 – Mock Interview

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	M Poojitha Reddy	Reg. No.:	192472352
Section / Batch:	2024	Date:	3-09-26

Code of Conduct

I M Poojitha Reddy - 192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Poojitha

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6

Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9
Integrated Case Study (Analytical & Problem Solving)	Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation	Q10

Annexure A – Mock Interview

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	2	2	2	2	2	10	100
2	2	2	2	2	2	10	
3	2	2	2	2	2	10	
4	2	2	2	2	2	10	

5	2	2	2	2	2	10	
6	2	2	2	2	2	10	
7	2	2	2	2	2	10	
8	2	2	2	2	2	10	
9	2	2	2	2	2	10	
10	2	2	2	2	2	10	

Faculty In-charge	Course Coordinator	Program Director
T Kumaragurubaran		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Analyze the reference resolution in: “Anita bought a laptop. She uses it every day.”

Answer:

The pronoun “**She**” refers to **Anita**, while “**it**” refers to the **laptop**. Resolving these references helps the system correctly understand the meaning of the discourse.

2. Analyze the coherence of: “It was raining heavily. Therefore, the match was cancelled.”

Answer:

The two sentences have a clear **cause-and-effect relationship**. Heavy rain is the cause, while cancellation of the match is the result. The word “**Therefore**” connects the two statements and creates coherence.

3. Identify the discourse relation in: “Ravi was tired, but he continued working.”

Answer:

The discourse relation is **contrast**. Ravi was tired, which would normally suggest stopping work, but he continued working. The conjunction “**but**” indicates the contrast.

4. Analyze the ambiguity in: “John told David that he was selected.”

Answer:

The pronoun “**he**” is ambiguous because it could refer to either **John or David**. Additional contextual information is required to determine who was selected.

5. Analyze the dialog act in: “Could you please open the window?”

Answer:

Although the sentence has the grammatical form of a **question**, its communicative function is a **request**. A dialogue system should understand that the speaker is asking someone to open the window.

6. Analyze the coherence of a paragraph containing unrelated sentences.

Answer:

If sentences discuss unrelated topics without logical connections or transitions, the paragraph **lacks coherence**. Readers may understand each sentence individually but may not understand the overall message.

7. Analyze the reference in: “The dog chased the cat because it was afraid.”

Answer:

The pronoun “**it**” is ambiguous because it may refer to either the **dog or the cat**. Semantic and contextual reasoning is required to determine which animal was afraid.

8. Analyze the discourse structure of an introduction, explanation, and conclusion.

Answer:

The **introduction** presents the main topic, the **explanation** develops the central ideas, and the **conclusion** summarizes the discussion. This structure provides a logical progression and improves readability.

9. Analyze why sentence ordering affects coherence.

Answer:

The order of sentences determines how information develops for the reader. A **logical sequence** helps readers follow events or arguments, while random ordering can make a meaningful text difficult to understand.

10. Analyze the dialog acts in: “Hello. How can I help you? I need information about admissions.”

Answer:

“Hello” is a **greeting**, while “How can I help you?” is an **offer of assistance**. The final statement is an **information request**. Thus, the conversation contains multiple dialog acts.

11. Analyze the role of context in interpreting the word “bank.”

Answer:

The word “bank” can refer to a **financial institution** or the **side of a river**. Surrounding words and discourse context help determine the correct meaning intended by the speaker.

12. Analyze the coherence relation in: “The road was blocked. As a result, we arrived late.”

Answer:

The phrase “As a result” signals a **cause-and-effect relationship**. The blocked road caused a delay, which resulted in the people arriving late.

13. Analyze how a chatbot should respond to: “I forgot my password.”

Answer:

The chatbot should identify the statement as a **request for account assistance**, even though the user does not explicitly ask a question. It should provide instructions for **resetting or recovering the password**.

14. Analyze the stages required to generate a weather report.

Answer:

The system first **selects important weather information**, such as temperature and rainfall. It then **organizes the information** and finally uses **surface realization** to generate fluent sentences.

15. Analyze the transfer approach for translating between two languages.

Answer:

The source sentence is first analyzed to identify its **grammatical and semantic structure**. This

structure is then transformed according to the rules of the target language before generating the final translated sentence.

16. Analyze the advantage of an interlingua over direct translation.

Answer:

An **interlingua** provides a common meaning representation that is independent of individual languages. This reduces the number of separate translation systems required when multiple languages need to communicate with one another.

17. Analyze why pronoun resolution is difficult in long documents.

Answer:

A pronoun may refer to an entity mentioned several sentences or paragraphs earlier. The system must track **entities, grammatical features, semantic information, and discourse context** to resolve the correct reference.

18. Analyze the metaphor: “Time is a thief.”

Answer:

The statement does not mean that time literally steals objects. It is a **metaphor** suggesting that time takes away opportunities, moments, and experiences from people's lives.

19. Analyze why literal translation may fail for idiomatic expressions.

Answer:

Idioms often have meanings that cannot be understood from their individual words. A **literal translation** may therefore produce meaningless or incorrect output. The translation system must identify the idiom and preserve its intended meaning.

20. Analyze the role of discourse planning in report generation.

Answer:

Discourse planning determines the order in which information should be presented. It ensures that important findings, supporting evidence, and conclusions are arranged in a meaningful and understandable structure.